

Physical Chemistry (II)

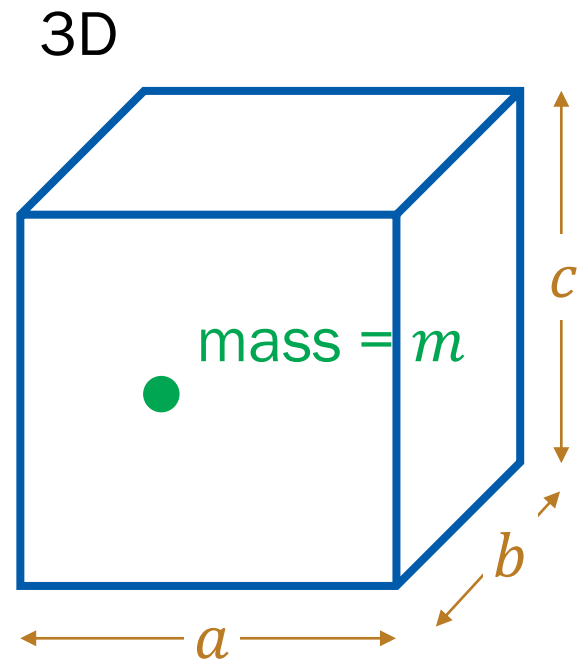
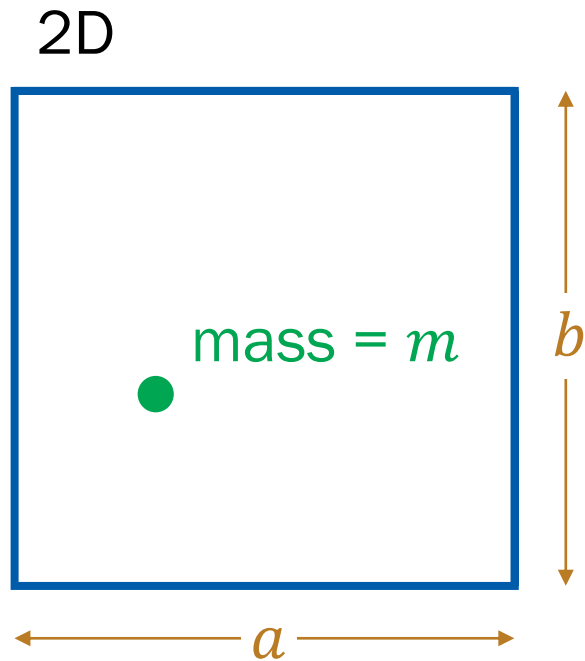
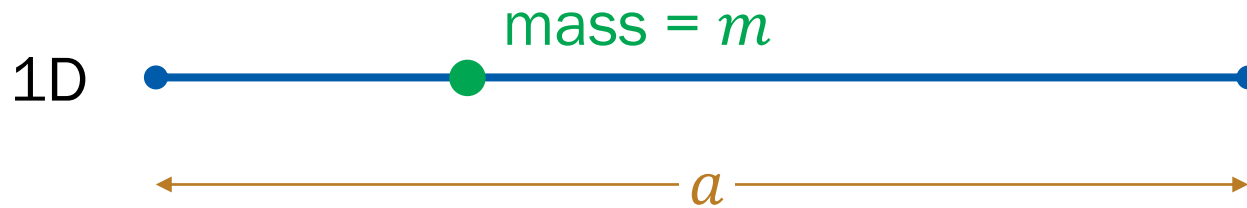
Lecture 06

Simple Harmonic Oscillator



Last class

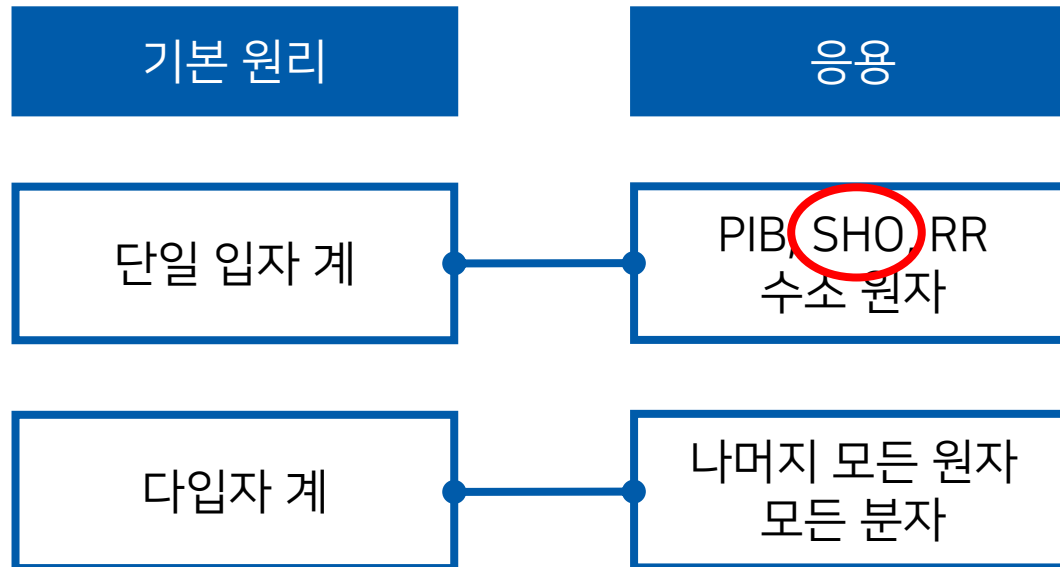
Particle in a Box



General Principles

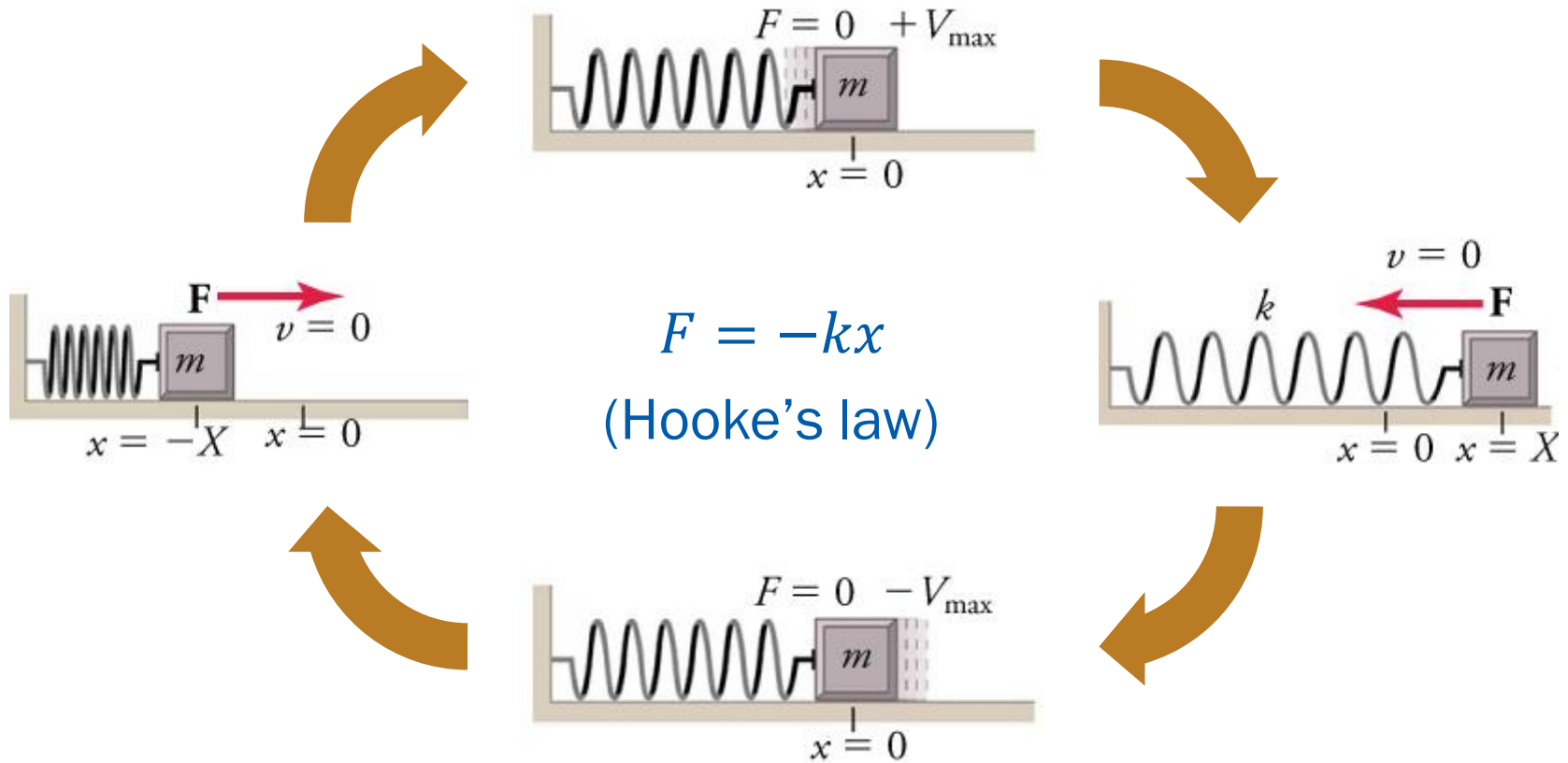
- Energy and wavefunctions
 - Transitions between energy levels → spectroscopy
 - Visualization of wavefunctions
- Quantum numbers
 - Number of quantum numbers = dimensionality
- Degeneracy
 - Only possible for 2 or more dimensions

Where Are We?



M 5.1

Simple Harmonic Oscillator (SHO)

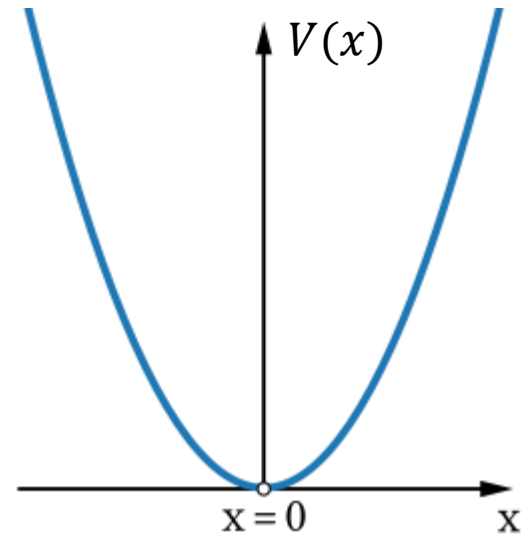


(Image: <https://openstax.org/books/physics/pages/5-5-simple-harmonic-motion>)

Potential

$$V(x) = \frac{1}{2} kx^2$$

$$F(x) = -\frac{dV}{dx} = -kx$$



(Image: <https://commons.wikimedia.org/wiki/File:Harmonic-oscillator-potential-energy-curve.svg>)

Quantum Mechanics

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \mathcal{V}(\hat{x}) \right] \psi(x) = E\psi(x)$$
$$\left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} kx^2 \right) \psi(x) = E\psi(x)$$

No simple method to solve this equation

(cf. operator method: RS 5.2, M Appendix 5)

But we “know” the solutions

Solutions

$$\left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} kx^2 \right) \psi(x) = E\psi(x)$$

- Quantum number: $v = 0, 1, 2, \dots$
- Energy

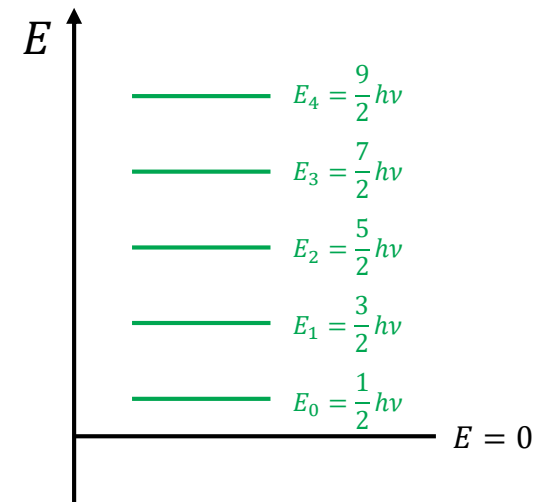
$$E_v = \frac{h}{2\pi} \left(\frac{k}{m} \right)^{1/2} \left(v + \frac{1}{2} \right) = h\nu \left(v + \frac{1}{2} \right)$$

Energy Levels

$$E_v = h\nu \left(v + \frac{1}{2} \right)$$
$$(v = 0, 1, 2, \dots)$$

- Zero-point energy

$$E_0 = \frac{1}{2} h\nu$$



- If $E = E_{\text{pot}} + E_{\text{k}} = \frac{1}{2} kx^2 + \frac{p^2}{2m} = 0$, both $x = p = 0$, which violates the uncertainty principle

Solutions

- Quantum number: $v = 0, 1, 2, \dots$
- Energy

$$E_v = \frac{h}{2\pi} \left(\frac{k}{m} \right)^{1/2} \left(v + \frac{1}{2} \right) = h\nu \left(v + \frac{1}{2} \right)$$

- Wavefunction

$$\psi_v(x) = \frac{(\alpha/\pi)^{1/4}}{(2^v v!)^{1/2}} \underbrace{H_v(\alpha^{1/2} x)}_{\text{Hermite polynomial}} e^{-\alpha x^2/2}$$

where $\alpha = \sqrt{mk}/\hbar$

Hermite Polynomials

$$H_n(\xi) = (-1)^n e^{\xi^2} \frac{d^n}{d\xi^n} e^{-\xi^2}$$

$$H_0(\xi) = 1$$

$$H_1(\xi) = 2\xi$$

$$H_2(\xi) = 4\xi^2 - 2$$

$$H_3(\xi) = 8\xi^3 - 12\xi$$

$\vdots$

Example: $\nu = 1$ State

$$\psi_\nu(x) = \frac{(\alpha/\pi)^{1/4}}{(2^\nu \nu!)^{1/2}} H_\nu(\alpha^{1/2} x) e^{-\alpha x^2/2}$$

For $\nu = 1$, $H_1(\xi) = 2\xi$ and

$$\psi_1(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} \frac{1}{2^{1/2}} \cdot (2\alpha^{1/2} x) e^{-\alpha x^2/2}$$

$$\psi_1(x) = \left(\frac{4\alpha^3}{\pi}\right)^{1/4} x e^{-\alpha x^2/2}$$

Eigenfunction?

$$\left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} kx^2 \right) \psi(x) = E\psi(x)$$

$$\psi_1(x) = \left(\frac{4\alpha^3}{\pi} \right)^{1/4} x e^{-\alpha x^2/2}$$

$$\text{LHS} = \left(\frac{4\alpha^3}{\pi} \right)^{1/4} \left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} (x e^{-\alpha x^2/2}) + \frac{1}{2} kx^3 e^{-\alpha x^2/2} \right]$$

Some Tedious Math

$$\text{LHS} = \left(\frac{4\alpha^3}{\pi} \right)^{1/4} \left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} (xe^{-\alpha x^2/2}) + \frac{1}{2} kx^3 e^{-\alpha x^2/2} \right]$$

$$\text{As } \frac{d^2}{dx^2} (xe^{-\alpha x^2/2}) = (\alpha^2 x^3 - 3\alpha x) e^{-\alpha x^2/2},$$

$$\text{LHS} = \left(\frac{4\alpha^3}{\pi} \right)^{1/4} \left[-\frac{\hbar^2}{2m} (\alpha^2 x^3 - 3\alpha x) e^{-\alpha x^2/2} + \frac{1}{2} kx^3 e^{-\alpha x^2/2} \right]$$

$$= \left(\frac{4\alpha^3}{\pi} \right)^{1/4} \left[\left(-\frac{\hbar^2 \alpha^2}{2m} + \frac{1}{2} k \right) x^3 e^{-\alpha x^2/2} + \frac{3\hbar^2 \alpha}{2m} x e^{-\alpha x^2/2} \right]$$

$$= 0 \quad (\because \alpha = \sqrt{mk}/\hbar)$$

Eigenvalue Equation!

$$\psi_1(x) = \left(\frac{4\alpha^3}{\pi}\right)^{1/4} x e^{-\alpha x^2/2}$$

$$\text{LHS} = \left(\frac{4\alpha^3}{\pi}\right)^{1/4} \frac{3\hbar^2\alpha}{2m} x e^{-\alpha x^2/2} = \frac{3\hbar^2\alpha}{2m} \psi_1(x)$$

Hence, we obtain $\hat{H}\psi_1(x) = E_1\psi_1(x)$ with

$$E_1 = \frac{3\hbar^2\alpha}{2m} = \frac{3\hbar^2}{2m} \frac{\sqrt{mk}}{\hbar} = \frac{3}{2} \hbar \sqrt{\frac{k}{m}} = \frac{3}{2} h\nu$$

RS 5.2

M 5.8

Normalization and Orthogonality

One can also show

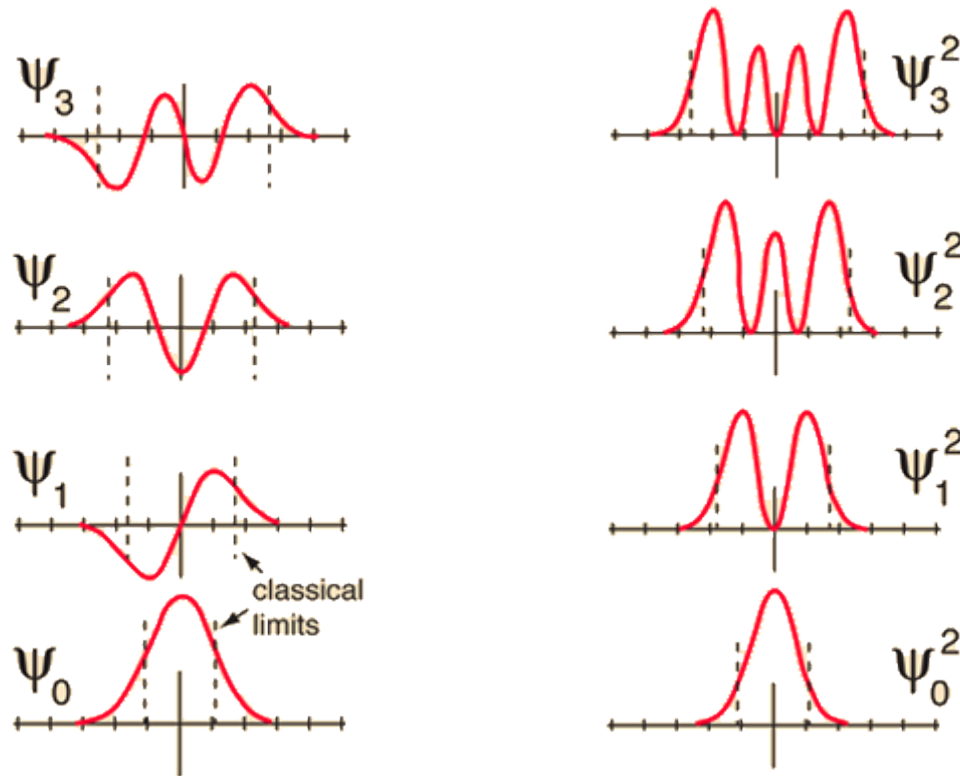
$$\int_{-\infty}^{\infty} |\psi_v(x)|^2 dx = 1$$

$$\int_{-\infty}^{\infty} \psi_v^*(x) \psi_u(x) dx = 0$$

for $u, v = 0, 1, 2, \dots$ and $u \neq v$

RS 5.2
M 5.8

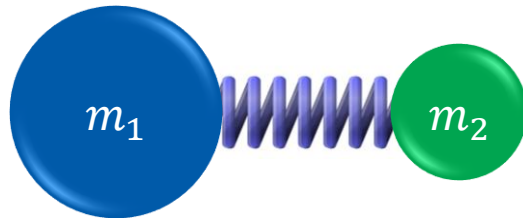
Wavefunctions



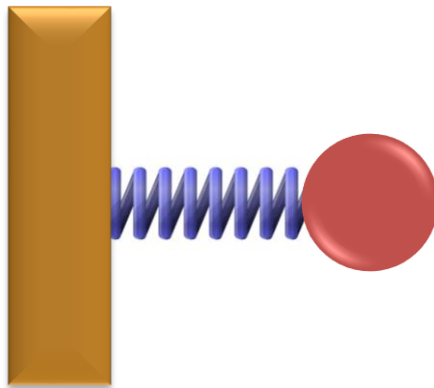
Tunneling!

(Image: <http://hyperphysics.phy-astr.gsu.edu/hbase/quantum/hosc7.html>)

Two Connected Bodies

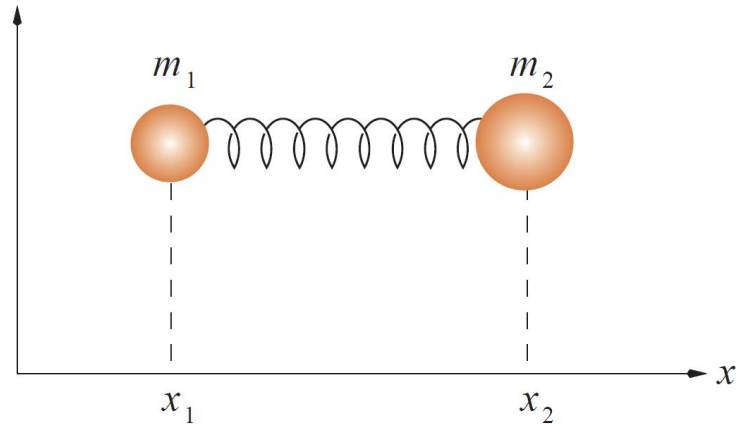


Two-body problem



One-body problem
(SHO)

Start from Classical Mechanics



From $ma = F$,

$$m_1 \frac{d^2 x_1}{dt^2} = k(x_2 - x_1 - l_0)$$

$$m_2 \frac{d^2 x_2}{dt^2} = -k(x_2 - x_1 - l_0)$$

(Image: McQuarrie, *Quantum Chemistry*, 2nd Ed., Fig. 5.4)

Addition of Equations

$$m_1 \frac{d^2 x_1}{dt^2} = k(x_2 - x_1 - l_0)$$

$$m_2 \frac{d^2 x_2}{dt^2} = -k(x_2 - x_1 - l_0)$$

Adding the two equations,

$$m_1 \frac{d^2 x_1}{dt^2} + m_2 \frac{d^2 x_2}{dt^2} = 0$$

$$\frac{d^2}{dt^2} (m_1 x_1 + m_2 x_2) = 0$$

Center-of-Mass Motion

$$\frac{d^2}{dt^2} (m_1 x_1 + m_2 x_2) = 0$$

Using

- Total mass $M = m_1 + m_2$
- Center-of-mass coordinate

$$X = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2},$$

$$M \frac{d^2 X}{dt^2} = 0 \text{ (constant motion)}$$

Subtraction of Equations

$$\frac{d^2 x_1}{dt^2} = \frac{k}{m_1} (x_2 - x_1 - l_0)$$

$$\frac{d^2 x_2}{dt^2} = -\frac{k}{m_2} (x_2 - x_1 - l_0)$$

Subtracting the first equation from the second,

$$\frac{d^2}{dt^2} (x_2 - x_1) = -k \left(\frac{1}{m_1} + \frac{1}{m_2} \right) (x_2 - x_1 - l_0)$$

$$\frac{d^2}{dt^2} (x_2 - x_1 - l_0) = -k \frac{m_1 + m_2}{m_1 m_2} (x_2 - x_1 - l_0)$$

Relative Motion

$$\frac{d^2}{dt^2} (x_2 - x_1 - l_0) = -k \frac{m_1 + m_2}{m_1 m_2} (x_2 - x_1 - l_0)$$

Define the relative coordinate

$$x = x_2 - x_1 - l_0$$

to obtain

$$\frac{m_1 m_2}{m_1 + m_2} \frac{d^2 x}{dt^2} = -kx$$

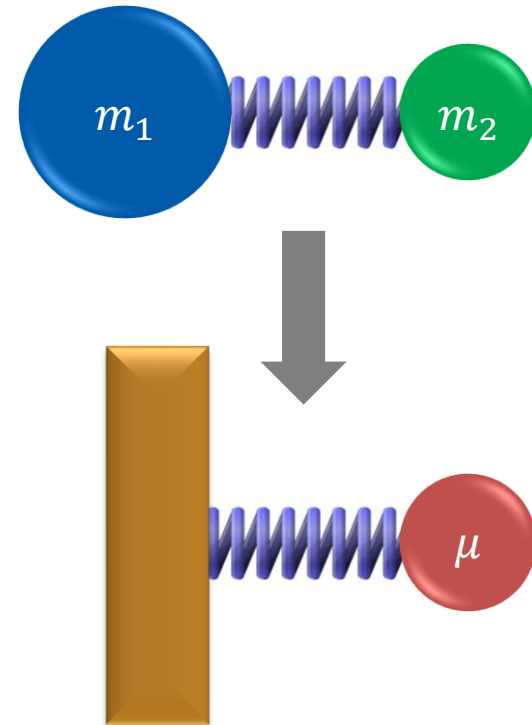
μ (reduced mass)

Summary

The motion between m_1 and m_2
is identical with the motion
between the fixed point and μ :

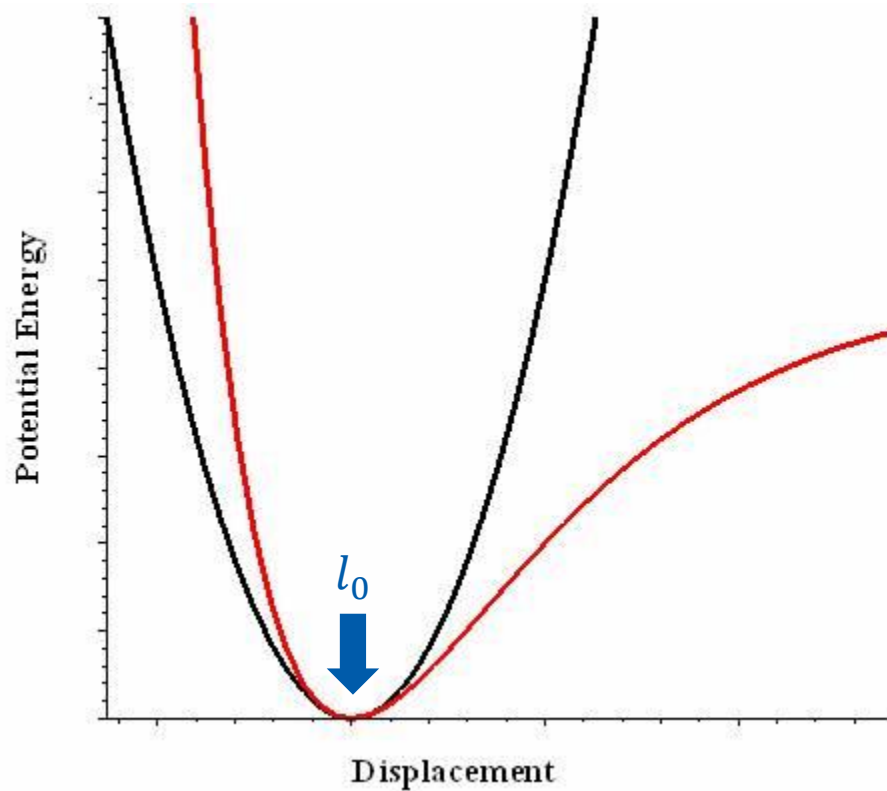
$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

(reduced mass)



RS 5.1
M 5.4

Approximation to Any Potential



(Image: <http://ueache.weebly.com/the-harmonic-and-anharmonic-oscillator.html>)

More Rigorous Proof

In the proximity of the minimum,

$$V(l) = V(l_0) + \left(\frac{dV}{dl}\right)_{l=l_0} (l - l_0) + \frac{1}{2!} \left(\frac{d^2V}{dl^2}\right)_{l=l_0} (l - l_0)^2 + \dots$$

At $l \approx l_0$, $(dV/dl) = 0$ and we can choose $V(l_0) = 0$. Hence,

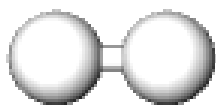
$$V(l) \approx \frac{1}{2!} \left(\frac{d^2V}{dl^2}\right)_{l=l_0} \frac{(l - l_0)^2}{x} + \dots$$

k

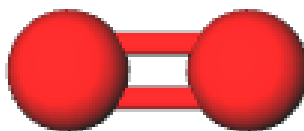
$$V(x) \approx \frac{1}{2} k x^2$$

RS 5.1
M 5.6

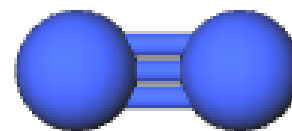
Model for Molecular Vibration



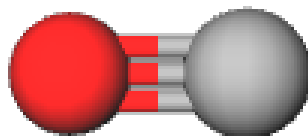
H₂



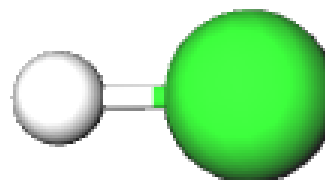
O₂



N₂



CO



HCl

Parameters

Molecule	k (N m ⁻¹)	l_0 (pm)
H ₂	510	74.14
D ₂	527	74.15
¹² C ¹⁶ O	1857	112.83
¹⁴ N ¹⁶ O	1550	115.08
¹⁴ N ¹⁴ N	2243	109.77
¹⁶ O ¹⁶ O	1142	120.75
H ¹⁹ F	920	91.68
H ³⁵ Cl	478	127.46
H ⁷⁹ Br	381	141.44
H ¹²⁷ I	291	160.92

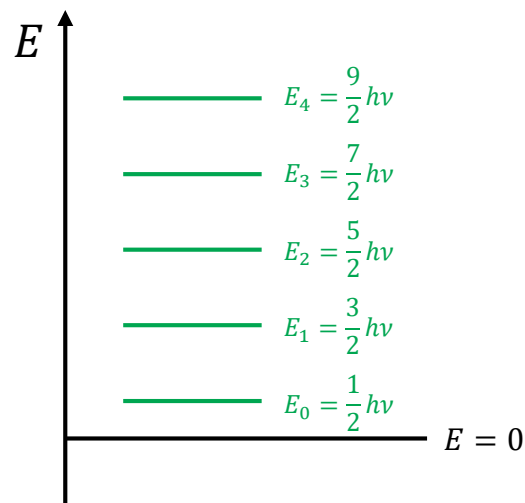
(Data: McQuarrie, *Quantum Chemistry*, 2nd Ed., Table 5.1)



Vibrational Energy Levels

$$E_v = h\nu \left(v + \frac{1}{2} \right), \quad \nu = \frac{1}{2\pi} \left(\frac{k}{\mu} \right)^{1/2}$$

$(v = 0, 1, 2, \dots)$



Thermal Excitations

$$P(E) \propto e^{-E/k_{\text{B}}T}, \quad E = E_v = h\nu(v + 1/2)$$

$$P_v \propto e^{-h\nu(v+1/2)/k_{\text{B}}T}$$

For typical diatomic molecules, $\nu \sim 10^{14}$ Hz

At 298 K, $h\nu/k_{\text{B}}T \sim 16.1$ and $P_v \propto e^{-16.1(v+0.5)} \propto e^{-16.1v}$

$$P_0 \approx 1, \quad P_1 \approx 10^{-7}, \quad P_2 \approx 10^{-14}, \quad \dots$$

At room temperature, most diatomic molecules are

in the ground vibrational state

(Transitions: $0 \rightarrow v$)

Selection Rules

- **Selection rules:** some transitions are allowed, and the others are prohibited by QM
- Selection rule 1 for SHO

$$\Delta v = \pm 1$$

- For absorption, $\Delta v = +1$ and

$$\Delta E = E_{v+1} - E_v = h\nu$$

- Only one line allowed: **fundamental vibrational frequency**
→ $\sim 10^{14}$ Hz (IR region) for diatomic molecules

(Modified) Example 5-4

The infrared spectrum of $^{75}\text{Br}^{19}\text{F}$ consists of an intense line at $26.3\text{ }\mu\text{m}$. Calculate the force constant of $^{75}\text{Br}^{19}\text{F}$.

$$\nu = \frac{1}{2\pi} \left(\frac{k}{\mu} \right)^{1/2} = \frac{c}{\lambda}$$

$$k = \mu \left(\frac{2\pi c}{\lambda} \right)^2 = \frac{m_1 m_2}{m_1 + m_2} \left(\frac{2\pi c}{\lambda} \right)^2$$

$$\mu = \frac{(75.0\text{ amu})(19.0\text{ amu})}{(75.0 + 19.0)\text{ amu}} (1.661 \times 10^{-27}\text{ kg amu}^{-1}) = 2.52 \times 10^{-26}\text{ kg}$$

$$\therefore k = (2.52 \times 10^{-26}\text{ kg}) \left(\frac{2\pi \times 2.998 \times 10^8\text{ m s}^{-1}}{26.3 \times 10^{-6}\text{ m}} \right)^2 = 129\text{ kg s}^{-2} = 129\text{ N m}^{-1}$$

Selection Rules

- **Selection rules:** some transitions are allowed, and the others are prohibited by QM
- Selection rule 1 for SHO

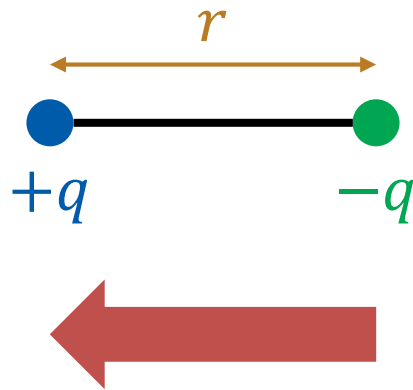
$$\Delta v = \pm 1$$

- Selection rule 2 for SHO

$$\frac{d\mu}{dx} \neq 0$$

“Dipole moment of the molecule must vary during vibration”

Dipole Moments



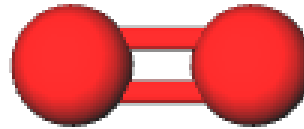
Dipole moment

$$\vec{\mu} = q\vec{r}$$

Units: Debye

$$1 \text{ D} = 3.34 \times 10^{-30} \text{ C m}$$

Homonuclear

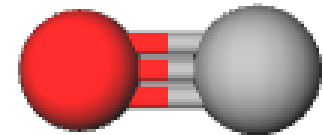


$$\frac{d\mu}{dx} = 0$$

($\mu = 0$ always)

NOT detectable

Heteronuclear



$$\frac{d\mu}{dx} \neq 0$$

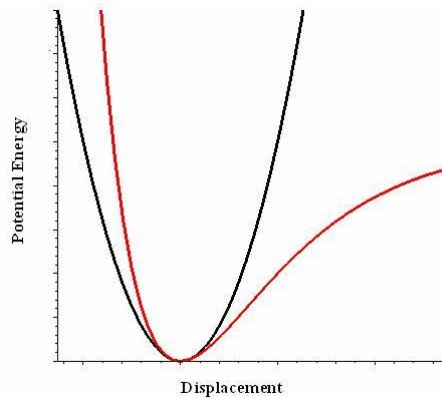
detectable

Anharmonicity

$$V(l) = \frac{1}{2!} \left(\frac{d^2 V}{dl^2} \right)_{l=l_0} (l - l_0)^2 + \frac{1}{3!} \left(\frac{d^3 V}{dl^3} \right)_{l=l_0} (l - l_0)^3 + \dots$$

$$V(x) = \frac{1}{2} k x^2 + \frac{1}{6} \gamma_3 x^3 + \frac{1}{24} \gamma_4 x^4 + \dots$$

anharmonicity



(Image: <http://ueache.weebly.com/the-harmonic-and-anharmonic-oscillator.html>)

Effects of Anharmonicity

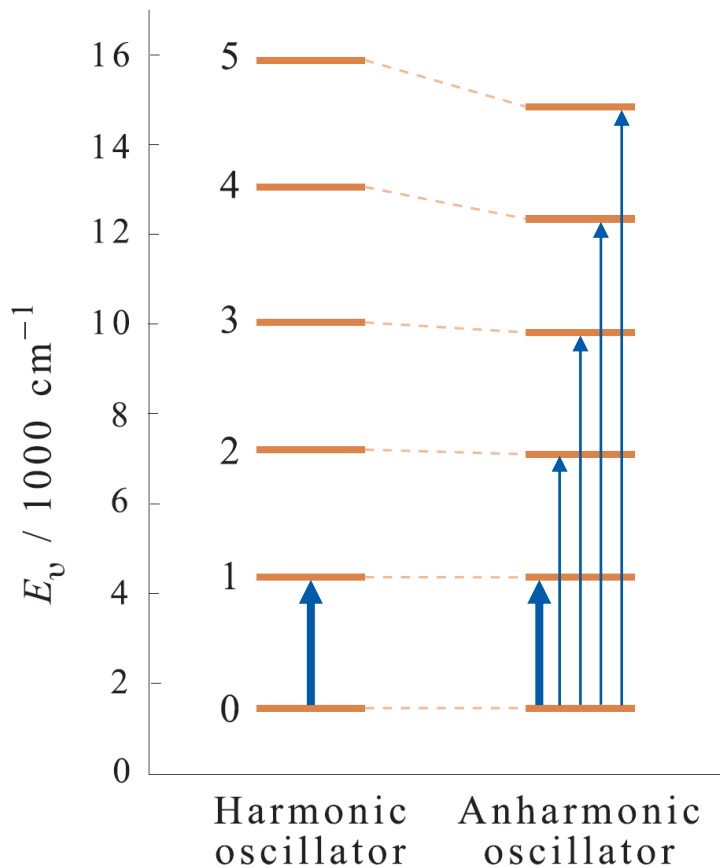
- Vibrational energy

$$E_v = h\nu \left(v + \frac{1}{2} \right) - \varepsilon_2 \left(v + \frac{1}{2} \right)^2 + \dots$$

- Selection rules

- $\Delta v = \pm 1$: major
- $\Delta v = \pm 2, \pm 3, \dots$: allowed (yet minor)

Effects of Anharmonicity



(Image: McQuarrie, *Quantum Chemistry*, 2nd Ed., Fig. 5.9)